

Steps

Reason

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| 4. $\forall x \neg (Q(x) \wedge R(x))$ | De Morgan's Law of quantifiers on 2 |
| 5. $\forall x (\neg Q(x) \vee \neg R(x))$ | 1st De Morgan's Law on 4 |
| 6. $\forall x (Q(x) \rightarrow \neg R(x))$ | Conditional-disjunction equivalence on 5 |
| 7. $Q(c) \rightarrow \neg R(c)$ | Universal instantiation on 6 |
| 8. $\neg R(c) \rightarrow \neg S(c)$ | Universal instantiation on 3 |
| 9. $Q(c) \rightarrow \neg S(c)$ | Hypothetical syllogism on 7, 8 |
| 10. $\neg(\neg S(c)) \rightarrow \neg Q(c)$ | Contrapositive on 9 |
| 11. $S(c) \rightarrow \neg Q(c)$ | Double-negation law on 10 |
| 12. $P(c) \rightarrow S(c)$ | Universal instantiation on 1 |
| 13. $P(c) \rightarrow \neg Q(c)$ | Hypothetical syllogism on 12, 11 |
| 14. $\forall x (P(x) \rightarrow \neg Q(x))$ | Universal generalization on 13 |

23) Identify the error or errors in this argument that supposedly shows that if $\exists x P(x) \wedge \exists x Q(x)$ is true then $\exists x (P(x) \wedge Q(x))$ is true.

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| 1. $\exists x P(x) \vee \exists x Q(x)$ | Premise |
| 2. $\exists x P(x)$ | Simplification from (1) |
| 3. $P(c)$ | Existential instantiation on 2 |
| 4. $\exists x Q(x)$ | Simplification from (1) |
| 5. $Q(c)$ | Existential instantiation from (4) |
| 6. $P(c) \wedge Q(c)$ | Conjunction from (3) and (5) |
| 7. $\exists x (P(x) \wedge Q(x))$ | Existential generalization |

Soln: In Step 1, the premise is wrong. It should be $\exists x P(x) \wedge \exists x Q(x)$.
• In Step 2, Step 4, Simplification is applied wrongly. We can conclude P, Q from $P \wedge Q$, but not from $P \vee Q$.
• In Step 5, it's not necessary that the element c in the domain which makes $P(c)$ True will also make $Q(c)$ True.